



Cambridge IGCSE™

COMPUTER SCIENCE

0478/23

Paper 2

May/June 2021

MARK SCHEME

Maximum Mark: 50

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2021 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **17** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Please note the following further points:

The words in **bold** in the mark scheme are important text that needs to be present, or some notion of it needs to be present. It does not have to be the exact word, but something close to the meaning.

If a word is underlined, this **exact** word must be present.

A single forward slash means this is an alternative word. A double forward slash means that this is an alternative mark point.

Ellipsis (...) on the end of one-mark point and the start of the next means that the candidate **cannot** get the second mark point without being awarded the first one. If a MP has ellipsis at the beginning, but there is no ellipsis on the MP before it, then this is just a follow-on sentence and **can** be awarded **without** the previous mark point.

Question	Answer	Marks
Section A		
1(a)(i)	One mark per bullet point • Constant NumberofOptions • Value 5 • Use Storing the maximum number of options (for the referendum)	3
1(a)(ii)	One mark per bullet point • Variable UniqueNumber • Use Storing/inputting the voter's unique identification number	2
1(a)(iii)	One mark for each valid mark point (Max 4) • Suitable name e.g. IdNumber • Suitable data type e.g. Integer • Suitable array length e.g. 170 (elements) • Suitable sample data e.g. 100 • Suitable use e.g. to store the unique numbers	4

Question	Answer	Marks
1(b)	Any six from: MP1 Initialisation of totalling variables MP2 Loop to cycle through all the data MP3 Totalling the votes for at least one of the options MP4 Totalling the votes for all five options MP5 Separate totals for students and staff MP6 Output to show one set of results MP7 Output to show more than one set of results with messages MP8 Comprehensive output to show student, staff and combined results, with messages Example answers Version 1 <pre> TotalAS ← 0; TotalBS ← 0; TotalCS ← 0; TotalDS ← 0; TotalsES ← 0; TotalAF ← 0; TotalBF ← 0; TotalCF ← 0; TotalDF ← 0; TotaleF ← 0 FOR Count ← 0 TO 150 TotalAS ← TotalAS + StudentA[Count] TotalBS ← TotalBS + StudentB[Count] TotalCS ← TotalCS + StudentC[Count] TotalDS ← TotalDS + StudentD[Count] Totales ← Totales + StudentE[Count] NEXT Count FOR Count = 0 TO 20 TotalAF ← TotalAF + StaffA [Count] TotalBF ← TotalBF + StaffA [Count] TotalCF ← TotalCF + StaffA [Count] TotalDF ← TotalDF + StaffA [Count] TotaleF ← TotaleF + StaffA [Count] NEXT Count OUTPUT "Option Students Staff Total" OUTPUT "A ", TotalAS, TotalAF, (TotalAS + TotalAF) OUTPUT "B ", TotalBS, TotalBF, (TotalBS + TotalBF) OUTPUT "C ", TotalCS, TotalCF, (TotalCS + TotalCF) OUTPUT "D ", TotalDS, TotalDF, (TotalDS + TotalDF) OUTPUT "E ", Totales, TotaleF, (Totales + TotaleF) </pre>	6

Question	Answer	Marks
1(b)	Version 2 TotalAS $\leftarrow$ 0; TotalBS $\leftarrow$ 0; TotalCS $\leftarrow$ 0; TotalDS $\leftarrow$ 0; TotalesES $\leftarrow$ 0; TotalAF $\leftarrow$ 0; TotalBF $\leftarrow$ 0; TotalCF $\leftarrow$ 0; TotalDF $\leftarrow$ 0; TotaleEF $\leftarrow$ 0 OUTPUT "Are you a student? (Y or N)" INPUT Student IF Student = "Y" THEN OUTPUT "Enter your number" INPUT Number IF Students[Number] = 0 THEN REPEAT OUTPUT "Enter preference for A" INPUT Vote UNTIL Vote $\geq$ 1 and Vote $\leq$ 5 TotalAS $\leftarrow$ TotalAS + Vote REPEAT OUTPUT "Enter preference for B" INPUT Vote UNTIL Vote $\geq$ 1 and Vote $\leq$ 5 TotalBS $\leftarrow$ TotalBS + Vote REPEAT OUTPUT "Enter preference for C" INPUT Vote UNTIL Vote $\geq$ 1 and Vote $\leq$ 5 TotalCS $\leftarrow$ TotalCS + Vote REPEAT OUTPUT "Enter preference for D" INPUT Vote UNTIL Vote $\geq$ 1 and Vote $\leq$ 5 TotalDS $\leftarrow$ TotalDS + Vote REPEAT OUTPUT "Enter preference for E" INPUT Vote UNTIL Vote $\geq$ 1 and Vote $\leq$ 5 TotalesES $\leftarrow$ TotalesES + Vote	

Question	Answer	Marks
1(b)	<pre> Student[Number] ← 1 ELSE OUTPUT "Enter your number" INPUT Number IF staff[number] = 0 THEN REPEAT OUTPUT "Enter preference for A" INPUT Vote UNTIL Vote >= 1 and Vote <= 5 TotalAS ← TotalAS + Vote REPEAT OUTPUT "Enter preference for B" INPUT Vote UNTIL Vote >= 1 and Vote <= 5 TotalBS ← TotalBS + Vote REPEAT OUTPUT "Enter preference for C" INPUT Vote UNTIL Vote >= 1 and Vote <= 5 TotalCS ← TotalCS + Vote REPEAT OUTPUT "Enter preference for D" INPUT Vote UNTIL Vote >= 1 and Vote <= 5 TotalDS ← TotalDS + Vote REPEAT OUTPUT "Enter preference for E" INPUT Vote UNTIL Vote >= 1 and Vote <= 5 TotaleS ← TotaleS + Vote Staff[Number] ← 1 ENDIF </pre>	

Question	Answer	Marks
1(b)	OUTPUT "Option Students Staff Total " OUTPUT "A ", TotalAS, TotalAF, (TotalAS + TotalAF) OUTPUT "B ", TotalBS, TotalBF, (TotalBS + TotalBF) OUTPUT "C ", TotalCS, TotalCF, (TotalCS + TotalCF) OUTPUT "D ", TotalDS, TotalDF, (TotalDS + TotalDF) OUTPUT "E ", TotalES, TotalEF, (TotalES + TotalEF)	
1(c)	Any two from: MP1 Change the maximum preferences constant to 6 MP2 Change the input prompt to show six options MP3 Change the loop counter MP4 Change the number of inputs for each person to six MP5 Add another array/variable to store/total the new option votes	2
1(d)	Explanation Any three from: MP1 Initialise count variables/array for each option MP2 Using a conditional statement to identify preference 1 in at least one option (in the input/stored data) MP3 ... identify preference 1 in all options MP4 ... and adding 1 to the appropriate option count in at least one option MP5 ... for students only MP6 ... using a loop to cycle through the whole array/set of inputs for each option	3

Question	Answer	Marks
Section B		
2	Three marks for three or four correct Two marks for two correct One mark for one correct Data Type Real String Integer Boolean Description Must be a whole number Must be one of two values May be any number May contain any combination of characters	3

Question	Answer	Marks
3	One mark per mark point for each piece of test data (3 × two marks) • Correct validation check name • Correct use identified Example answers ld27@cambridgeuniversity.com • Validation check name Length (check) • Use Counts the number of characters in the data to make sure it isn't too long (max length 320 characters). 2021 • Validation check name Range (check) • Use Checks that the number entered fits within given parameters Ericson-Bower • Validation check name Type (check) • Use Checks the type of data entered (in this case) to make sure no numbers are present	6

Question	Answer	Marks
4(a)	One mark for error identified and suggested correction Line 3 – should be <code>FullScore ← 0</code> Line 7 – should be <code>FullScore ← FullScore + Score</code> Line 8 – should be <code>NEXT Allow ENDFOR //</code> alternatively Line 5 could be <code>REPEAT</code> with <code>StoreLoop ← 0</code> just above it and <code>StoreLoop ← StoreLoop + 1</code> between lines 7 and 8. Line 11 – should be <code>INPUT Another</code> Correct Algorithm 1 <pre> 1 Count ← 0 2 REPEAT 3 FullScore ← 0 4 INPUT Number 5 FOR StoreLoop ← 1 TO Number 6 INPUT Score 7 FullScore ← FullScore + Score 8 NEXT 9 OUTPUT "The full score is ", FullScore 10 OUTPUT "Another set of scores (Y or N)?" 11 INPUT Another 12 IF Another = "N" 13 THEN 14 Count ← 1 15 ENDIF 16 UNTIL Count = 1 </pre>	4

Question	Answer	Marks
4(a)	Correct Algorithm 2 <pre> 1 Count ← 0 2 REPEAT 3 FullScore ← 0 4 INPUT Number 5 StoreLoop ← 0 6 REPEAT 7 INPUT Score 8 FullScore ← FullScore + Score 9 StoreLoop ← StoreLoop + 1 10 UNTIL StoreLoop = Number 11 OUTPUT "The full score is ", FullScore 12 OUTPUT "Another set of scores (Y or N)?" 13 INPUT Another 14 IF Another = "N" 15 THEN 16 Count ← 1 17 ENDIF 18 UNTIL Count = 1 </pre>	

Question	Answer	Marks
4(b)	One mark per mark point (Max 4) MP1 After line 6 // replace line 6 MP2 ScoreArray[StoreLoop] ← Score // INPUT ScoreArray[StoreLoop] MP3 between lines 8 and 10 MP4 AverageScore ← FullScore/Number MP5 OUTPUT "The average score is ", AverageScore Example correct algorithm for reference from part 4(a) <pre> 1 Count ← 0 2 REPEAT 3 FullScore ← 0 4 INPUT Number 5 FOR StoreLoop ← 1 TO Number 6 INPUT Score 7 FullScore ← FullScore + Score 8 NEXT 9 OUTPUT "The full score is ", FullScore 10 OUTPUT "Another set of scores (Y or N)?" 11 INPUT Another 12 IF Another = "N" 13 THEN 14 Count ← 1 15 ENDIF 16 UNTIL Count = 1 </pre>	4

Question	Answer	Marks																																																																																															
5(a)	One mark for each correct column <table><tr><th>Op</th><th>Value1</th><th>Value2</th><th>Ans</th><th>OUTPUT</th></tr><tr><td>1</td><td></td><td></td><td></td><td></td></tr><tr><td></td><td>87</td><td></td><td></td><td></td></tr><tr><td></td><td></td><td>14</td><td>101</td><td></td></tr><tr><td>3</td><td></td><td></td><td></td><td></td></tr><tr><td></td><td>2</td><td></td><td></td><td></td></tr><tr><td></td><td></td><td>30</td><td>60</td><td></td></tr><tr><td>5</td><td></td><td></td><td></td><td></td></tr><tr><td></td><td>10</td><td></td><td></td><td></td></tr><tr><td></td><td></td><td>6</td><td></td><td>Input Error</td></tr><tr><td>4</td><td></td><td></td><td></td><td></td></tr><tr><td></td><td>10</td><td></td><td></td><td></td></tr><tr><td></td><td></td><td>2</td><td>5</td><td></td></tr><tr><td>0</td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td></tr></table>	Op	Value1	Value2	Ans	OUTPUT	1						87						14	101		3						2						30	60		5						10						6		Input Error	4						10						2	5		0																														5
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Question	Answer	Marks
5(b)	To work as a calculator // to add, subtract, multiply or divide a pair of numbers	1
5(c)	To output/store the result/the value of <code>Ans</code> // Adding prompts for data entry.	1

Question	Answer	Marks							
6(a)	Two marks for six correct field names One mark for at least three correct field names <table><tr><th>Field name</th></tr><tr><td>LicenceNo</td></tr><tr><td>Mileage</td></tr><tr><td>TyreFLft</td></tr><tr><td>TyreFRgt</td></tr><tr><td>TyreRLft</td></tr><tr><td>TyreRRgt</td></tr></table>	Field name	LicenceNo	Mileage	TyreFLft	TyreFRgt	TyreRLft	TyreRRgt	2
Field name									
LicenceNo									
Mileage									
TyreFLft									
TyreFRgt									
TyreRLft									
TyreRRgt									

Question	Answer	Marks																																										
6(b)	One mark for correct fieldnames One mark for correct table names and show fields One mark for correct sort One mark for correct search criteria in all columns <table><tr><td>Field:</td><td>LicenceNo</td><td>Mileage</td><td>TyreFLft</td><td>TyreFRgt</td><td>TyreRLft</td><td>TyreRRgt</td></tr><tr><td>Table:</td><td>TREAD</td><td>TREAD</td><td>TREAD</td><td>TREAD</td><td>TREAD</td><td>TREAD</td></tr><tr><td>Sort:</td><td>Ascending</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>Show:</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td><td>☒</td></tr><tr><td>Criteria:</td><td></td><td></td><td><2</td><td><2</td><td><2</td><td><2</td></tr><tr><td>or:</td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>	Field:	LicenceNo	Mileage	TyreFLft	TyreFRgt	TyreRLft	TyreRRgt	Table:	TREAD	TREAD	TREAD	TREAD	TREAD	TREAD	Sort:	Ascending						Show:	☒	☒	☒	☒	☒	☒	Criteria:			<2	<2	<2	<2	or:							4
Field:	LicenceNo	Mileage	TyreFLft	TyreFRgt	TyreRLft	TyreRRgt																																						
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